

Tree Log Detection Project

This project aims to develop a machine learning model for detecting and counting tree logs using simulated LiDAR point cloud data.

Project Checklist

- ☒ Initial project setup
- ☒ Basic Blender scene generation
- ☒ Single log type implementation
- ☐ Physics simulation tuning
- ☐ LiDAR sensor simulation
- ☐ Dataset generation pipeline
- ☐ Initial dataset creation (10,000 scenes)
- ☐ Neural network architecture design
- ☐ Model training pipeline
- ☐ Model evaluation and testing
- ☐ Documentation completion

Project Phases

Phase 1: Data Generation using Blender

- Automated scene generation using Blender's Python API
- Random placement of tree log objects in the scene
- Physics simulation for realistic log placement

Phase 2: LiDAR Data Generation

- Simulate LiDAR sensor to generate point cloud data
- Generate large-scale dataset (target: 10,000 scenes)
- Dataset will contain:
 - Point cloud data for each scene
 - Ground truth log count per scene

Phase 3: Machine Learning Model

- Develop neural network architecture for log detection
- Train model using generated dataset
- Goal: Accurate prediction of log count from point cloud data